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Section 5.3: Database Structures

This section focuses on databases. You will see examples of databases and understand their purpose.

We will discuss **terminologies** (words) associated with databases and also the different ways in which they are **structured**.

You will also take a look at the different ways in which databases **organise data** to make it easier to search through.

Key Concepts of this section:

- # Understand database key-words including **record**, **file**, **table**, **field**, **data** and **key-field**.
- # Understand the ways in which databases are **organised** and **structured**.
- # Understand how databases can be **sorted** in both **ascending** and **descending** order.

What is a Database?

- # A database is **store of data** that has been **organised** (sorted) in some way.
- # Databases of data are held on a computer.

Example of a database with stored data.

The database below shows you an example of stored medical patients data:



Key Words:

Computer Data, Stored, Organised

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06

NOTE!

The database has been sorted into order of **Patient Id Number**.

Breakdown of the medical patients database.

- # Medical Centres, Doctors Surgeries and Hospitals hold lots of data about patients.
- # The image below shows you examples of the data they might hold. Notice how the information stored about each patient is similar (Name, Gender, D.o.B etc).

REMEMBER!

Databases can store data in an organised way. This makes **searching, updating** and **outputting** the data very easy.



Patient Id: 134
Name: Jeff
D.o.B: 4-Jul-1993
Gender: Male
Phone: 7876453
Doctor Id: 01
Doctor: Dr Hyde
Room: 03



Patient Id: 178
Name: David
D.o.B: 8-Feb-1987
Gender: Male
Phone: 8635467
Doctor Id: 02
Doctor: Dr Jekyll
Room: 06



Patient Id: 198
Name: Lisa
D.o.B: 18-Dec-1979
Gender: Female
Phone: 7498735
Doctor Id: 01
Doctor: Dr Hyde
Room: 03



Patient Id: 210
Name: Frank
D.o.B: 29-Apr-1983
Gender: Male
Phone: 7943521
Doctor Id: 01
Doctor: Dr Hyde
Room: 03



Patient Id: 258
Name: Rachel
D.o.B: 8-Feb-1987
Gender: Female
Phone: 8367242
Doctor Id: 02
Doctor: Dr Jekyll
Room: 06

Database Structures (Files/Tables, Records, Fields, Data)



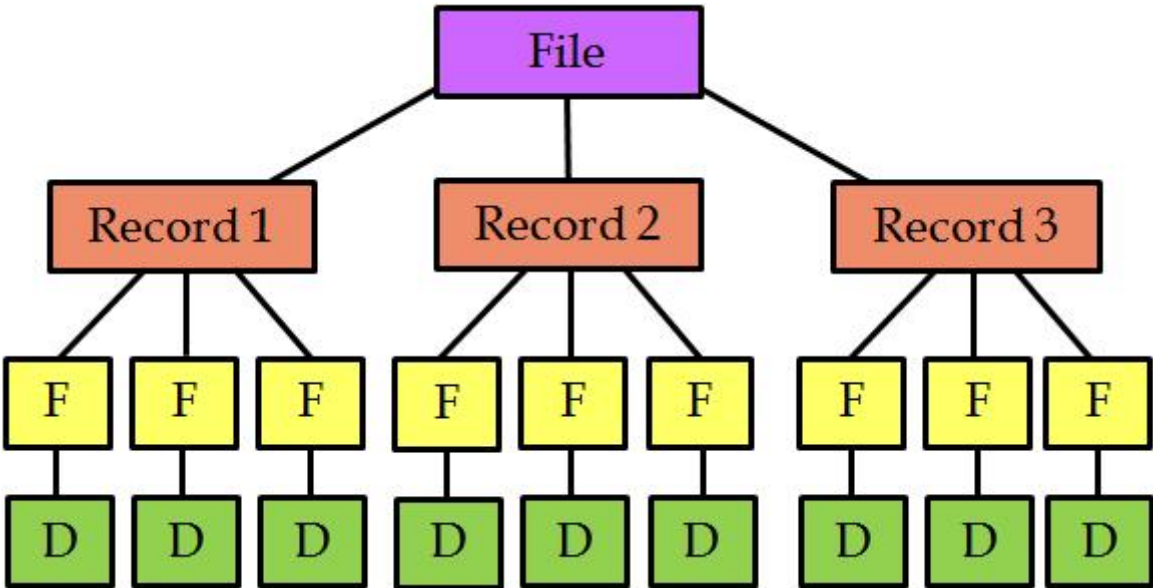
Key Words:

Database Structure,
File, Table, Record,
Field, Data

- # Databases are so useful because they **store data in a structured way**.
- # When data is structured (organised) it can be **manipulated easily** and then **output in different ways**
- # Databases consist of the following:
 1. **Files / Tables**
 2. **Records**
 3. **Fields**
 4. **Data**

Example of the structure of a database.

The image below shows how a database is structured into **Files/Tables, Records, Fields** and **Data**:



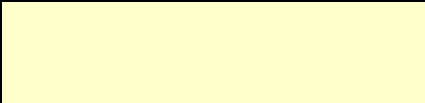
NOTE!
Files / Tables hold Records.
Records consist of Fields.
Fields hold Data.

Files/Tables, Records, Fields and Data are explained in more detail below

1 Files/Tables

- # A **collection of records and fields** is called a **file**.
- # Database files are viewed as **tables**. The medical patients database is holding its data in a table. Lets have another look at it.
- # The patients table is made up of **8 columns** and **5 rows**. **Column headings** are known as **fields** and **one complete row** is a full **record** of data.

NOTE!
Tables allow you to see all of the data at once rather than individually.



Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06

NOTE!

Tables are made up of **columns** and **rows**.

In the example to the left the **green box** shows a **column** and the **red box** shows a **row**.

2 Database Records

One **single row** of a database table represents **ONE Record**:

For example:- I have highted in **red**, record number 2 in our patient database:

NOTE!

In this particular example there are **5 records**.

One
Record

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06

What is a record?

A database record shows **all the data** about **one person or object**.

For example:- the image above shows the entire record for the patient named David.

The record contains the following information:



Patient Id: 178
Name: David
D.o.B: 8-Feb-1987
Gender: Male
Phone: 8635467
Doctor Id: 02
Doctor: Dr Jekyll
Room: 06

NOTE!

The headings for each piece of information (Name, Gender etc) are known as **fields**.

The records for the other patients are shown below.



Record 1

Patient Id: 134
Name: Jeff
D.o.B: 4-Jul-1993
Gender: Male
Phone: 7876453
Doctor Id: 01
Doctor: Dr Hyde
Room: 03

Record 3

Patient Id: 198
Name: Lisa
D.o.B: 18-Dec-1979
Gender: Female
Phone: 7498735
Doctor Id: 01
Doctor: Dr Hyde
Room: 03

Record 4

Patient Id: 210
Name: Frank
D.o.B: 29-Apr-1983
Gender: Male
Phone: 7943521
Doctor Id: 01
Doctor: Dr Hyde
Room: 03

Record 5

Patient Id: 258
Name: Rachel
D.o.B: 8-Feb-1987
Gender: Female
Phone: 8367242
Doctor Id: 02
Doctor: Dr Jekyll
Room: 06

NOTE!

Each record has the **same data structure** (Id, Name, Gender etc).

The only things that change are the individual pieces of information (Actual Names, Actual Genders etc)

3 Database Fields

Each **cell of a table** indicates a **single field**. A single field can **hold one piece of data**.

Fields are given **field names** so we know what kind of data they hold. Field names are found at the **top of each column**.

For example:- I have highlighted in **green** all of the field names in the patient database and in **red** a single field holding individual data:

NOTE!

In this particular example fields include **Name, D.o.B, Gender, Doctor** etc.

Field
Names

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
198	Lisa	18-Dec-1979	Female	48735	01	Dr Hyde	03
210	Frank				01	Dr Hyde	03
258	Rachel				02	Dr Jekyll	06

Single Field holding the Phone
Number of Patient 178

Fields are made up of 2 parts:-

Part 1: The Field

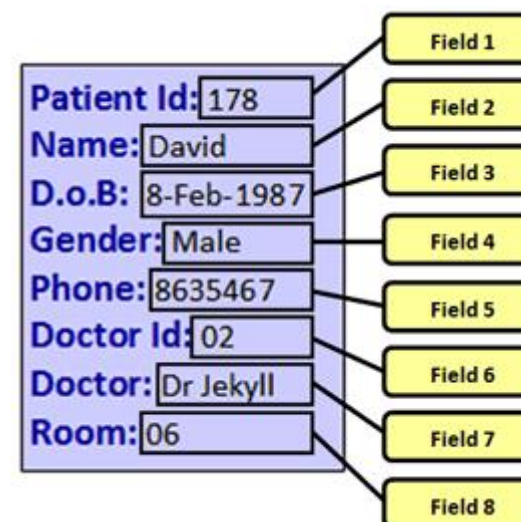
Part 2: The Field Name

What is a field?

Database fields **hold/store** the **single items of data** that make up a record.

For example:- each of our records consist of 8 fields:

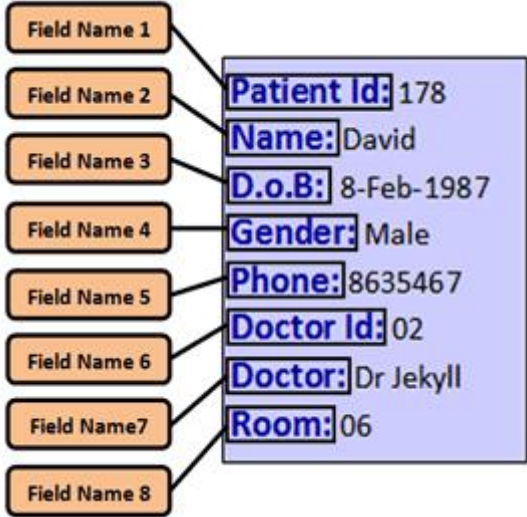
1. Patient Id
2. Name
3. D.o.B
4. Gender
5. Phone
6. Doctor Id
7. Doctor
8. Room



What is a field name?

Fields names are used to describe the data that is to be stored inside the field.

For example:- D.o.B: is the field name used to describe the **Date of Birth field**. It helps the user decide what data to enter.



4 Database Data

What is data?

Data is the **individual pieces of information** that get stored in a field.

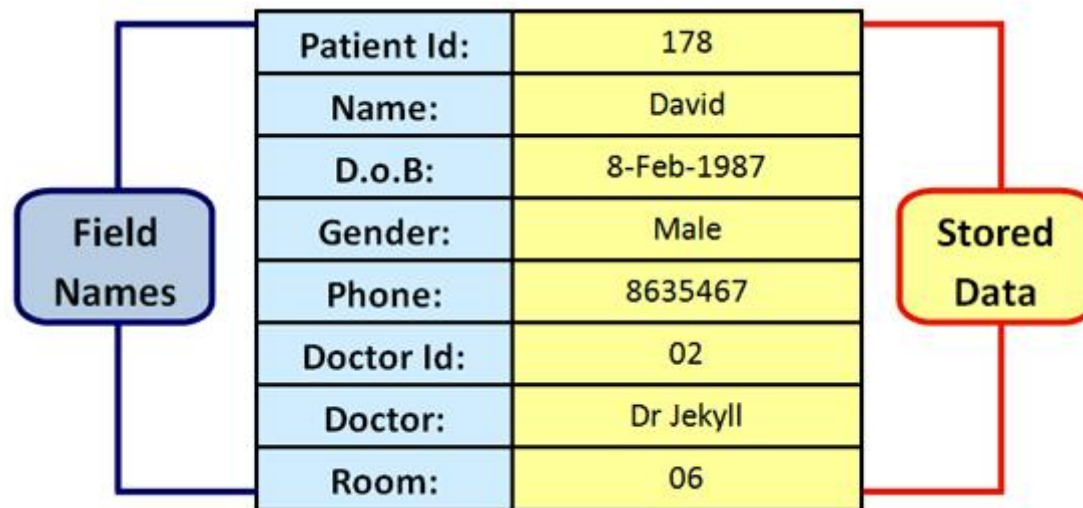
For example:

In the table below I have listed each field in our patient database and an example of data that could be stored inside it (E.g. **'David'** is stored in the **'Name' field**):

REMEMBER!

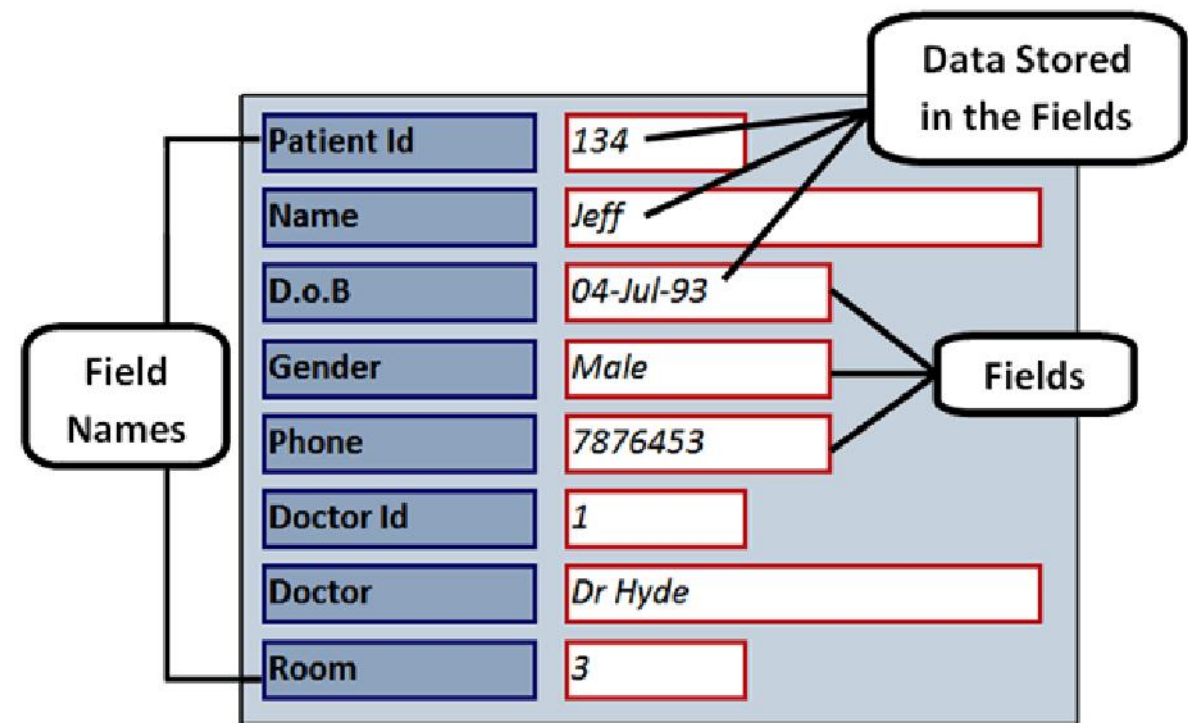
The data that is stored inside fields can **change from record to record**.

For example: The names **'David'**, **'Lisa'**, **'Frank'**, **'Jeff'** and **'Rachel'** can all be stored in the **Name Field**.



Fields, Field Names and Data example

The image to the right shows you a working example of **fields**, **field names** and **data** being used on an actual database:



Sorting Data (organising)



Key Words:

Ascending, Descending

As mentioned before, one of the uses of storing data on a database is that they allow data to be **sorted very easily**.

Data can be sorted in two ways:

1. **Ascending sorting**
2. **Descending sorting**

1 Ascending Sorting (lowest to highest)

Sorting data in '**ascending order**' means to start with the **lowest value** and move through it in order **until you reach the highest value**.

Think of these 3 examples:

1. **Text data** would be sorted from **A to Z**
2. **Number data** would be sorted from **0 - whatever** (or even negative numbers)
3. **Dates** would be sorted from the **earliest dates** to the **most recent**.

Take a look at our patients database again. At the moment it is sorted in **Ascending Order** of the **Patient Id** field:

NOTE!

Only Patient Id is in order.
None of the other fields
are sorted at all.

Ascending Order Example

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06

Lowest number
at the top

Highest number
at the bottom

2 Descending Sorting (highest to lowest)

Sorting data in '**descending order**' means to start with the **highest value** and move through it in order **until you reach the lowest value**.

Think of these 3 examples:

1. **Text data** would be sorted from **Z to A**
2. **Number data** would be sorted from **whatever - 0**
3. **Dates** would be sorted from the **most recent date** to the **earliest**.

Lets sort the patient database in **Descending Order** of the **Patient Id** field:

NOTE!

Only Patient Id is in order.
None of the other fields
are sorted at all.

Descending Order Example

Highest number
at the top

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06
210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03

Lowest number
at the bottom



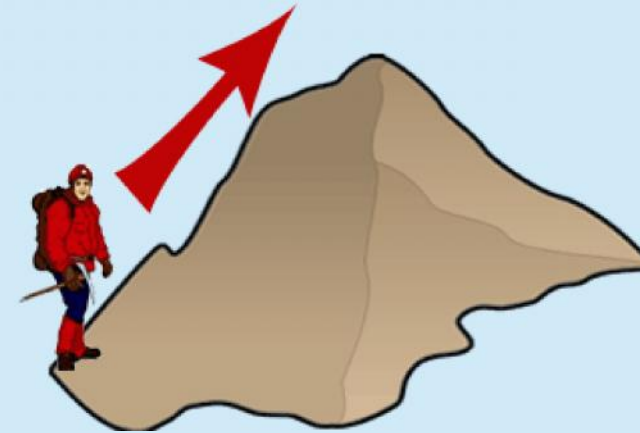
How to remember the difference between ascending and descending.

- 1.** Climbers who '**ascend**' a mountain start from the **bottom** and climb to the **top**.
- 2.** Climbers who '**descend**' a mountain start at the **top** and climb to the **bottom**.

'Descending' a mountain
(high point to the low point)



'Ascending' a mountain
(low point to the high point)



Key Fields (unique data)

- # Key fields are **individual pieces of data** that are **unique** (only appear once).
- # Key fields are used to **differentiate one record from another**.

For example:- In our patients database the key field is the **Patient Id**. Every patient has a **different Id Number**. This is vital so we can **identify each patient uniquely and correctly**. Imagine if the doctor mixed two patients up and gave them the wrong medicine!!



Key Words:

Key Field , Unique, Non-Repeating.

The **Patient Id** Number Field contains **non-repeating** data that

Patient Database Key Field

identifies individual records. We call this the **Key Field**.

	Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
Unique	258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06
Unique	210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
Unique	198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
Unique	178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
Unique	134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03

Why is Patient Id suitable for the Key Field?

Only the Patient Id is suitable for use as the unique key field. Remember the key field **MUST contain only data that CAN NEVER REPEAT**. Look at the table below and think about it:

Name ❌ Two patients can have the same name.

D.o.B ❌ Two patients can have the same birthday (Like Rachel and David in our table).

Gender ❌ Two patients can be the same gender.

Phone ❌ Although the phone numbers are all unique, phone numbers can be changed or even swapped. This makes them unsuitable for key fields.

Doctor Id ❌ The doctor Id can not be used to uniquely identify a patient..... just each doctor.

Doctor ❌ Doctor's names cannot be used to identify each patient as two or more patients can be assigned the same doctor (like in our table).

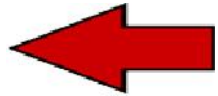
Room ❌ Multiple patients are sent to the same room. This is not a good key field.

Patient Id ✅ Each patient is given a different Id number which they keep for life. It is used to identify their medical history and any medicines they need.

Activity!

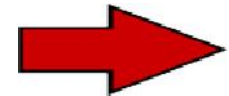
[Database Structures - Task](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete all tasks on the sheet.



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